

Range

Scenario: Lane Change in front of the Ego car on US101 Highway - Original

Trajectory Prediction Model		CS-LSTM										STDAN									
Manouver		Right Lane Change					Left Lane Change					Right Lane Change					Left Lane Change				
Sensor Parameter		Range					Range					Range					Range				
Sensor Modality	Comparison Metrics	50m	75m	100m	125m	150m	50m	75m	100m	125m	150m	50m	75m	100m	125m	150m	50m	75m	100m	125m	150m
Radar	Pearson Coeff	0.9889	0.9906	0.9933	0.989	0.9884	0.9926	0.9927	0.9954	0.994	0.9945	0.9997	0.9997	0.9996	0.9997	0.9996	0.9924	0.9948	0.9946	0.9935	0.9939
	Max Error	0.9383	0.8076	0.7117	0.8106	0.8468	1.4872	1.5556	1.0724	1.238	1.3606	0.3175	0.3257	0.2563	0.4093	0.3182	1.1622	1.1292	1.0994	1.1962	1.0962
	MAE	0.3072	0.2568	0.2624	0.2606	0.2558	0.7865	0.869	0.5165	0.6954	0.7528	0.2244	0.2251	0.1885	0.2668	0.2284	0.752	0.7417	0.7082	0.7689	0.7283
	AUC	30.6029	29.4014	29.6037	29.5104	29.3635	42.2551	44.3038	35.7392	40.1254	41.4897	51.9927	52.0038	51.1398	53.0065	52.0912	54.1566	53.8773	53.0634	54.5484	53.5676
	SNR	9.4996	11.006	11.2975	10.7896	10.8227	2.7824	1.8411	6.2582	3.9392	3.052	20.1331	20.0704	21.7122	18.4509	19.9971	6.7752	6.8917	7.2873	6.549	7.0976

Scenario: Lane Change in front of the Lead car on US101 Highway - Original

Trajectory Prediction Model		CS-LSTM										STDAN									
Manouver		Right Lane Change					Left Lane Change					Right Lane Change					Left Lane Change				
Sensor Parameter		Range					Range					Range					Range				
Sensor Modality	Comparison Metrics	50m	75m	100m	125m	150m	50m	75m	100m	125m	150m	50m	75m	100m	125m	150m	50m	75m	100m	125m	150m
Radar	Pearson Coeff	0.9903	0.9901	0.99	0.9899	0.9837	0.9918	0.9924	0.989	0.9911	0.9907	0.9978	0.9972	0.9976	0.9975	0.9974	0.9925	0.9914	0.9916	0.9945	0.9914
	Max Error	0.9643	1.1493	1.1733	1.2361	1.415	2.2544	1.5499	1.6236	2.3211	2.1627	1.3742	1.529	1.5608	1.5607	1.5457	1.559	1.4343	1.4508	1.7298	1.5027
	MAE	0.3962	0.5092	0.4961	0.5583	0.5148	1.3007	0.9032	1.0151	1.1025	1.3277	0.8582	0.9376	0.9512	0.945	0.9447	0.9426	0.8543	0.8693	1.031	0.9026
	AUC	32.6655	35.403	35.0335	36.5572	35.3879	54.4336	44.8341	47.6155	49.4288	55.1474	55.8677	57.8144	58.1191	57.9834	57.9784	62.1856	60.0831	60.4402	64.2827	61.2426
	SNR	7.9707	6.0408	6.1497	5.2799	5.3113	-1.8599	1.5518	0.5547	-0.7546	-1.9949	5.7893	4.9584	4.8149	4.8392	4.8495	5.6646	6.5423	6.4002	4.8245	6.0537

Scenario: Deceleartion on US101 Highway - Original

Trajectory Prediction Model		CS-LSTM					STDAN				
Manouver		Deceleartion					Deceleartion				
Sensor Parameter		Range					Range				
Sensor Modality	Comparison Metrics	50m	75m	100m	125m	150m	50m	75m	100m	125m	150m
Radar	Pearson Coeff	0.9972	0.9969	0.9967	0.9966	0.9975	0.9999	0.9999	0.9999	0.9999	0.9999
	Max Error	0.22	0.1752	0.2971	0.4404	0.1276	0.2193	0.2517	0.2619	0.2428	0.2477
	MAE	0.0505	0.0485	0.0983	0.169	0.0397	0.1214	0.1355	0.1414	0.1344	0.1343
	AUC	24.9635	24.5075	26.2472	27.9437	23.981	37.7946	38.1314	38.2736	38.1091	38.1031
	SNR	24.6397	26.169	19.9566	15.6924	27.9134	22.2947	21.2799	20.9019	21.3916	21.3594

HFOV

a) Scenario: Lane Change in front of the Ego car on US101 Highway - Original																									
Trajectory Prediction Model		CS-LSTM											STDAN												
Manouver		Right Lane change						Left Lane change						Right Lane change						Left Lane change					
Sensor Parameter		HFOV						HFOV						HFOV						HFOV					
Sensor Modality	Comparison Metrics	30deg	60deg	90deg	120deg	150deg	180deg	30deg	60deg	90deg	120deg	150deg	180deg	30deg	60deg	90deg	120deg	150deg	180deg	30deg	60deg	90deg	120deg	150deg	180deg
Radar	Pearson Coeff	0.9969	0.9886	0.9915	0.9901	0.9893	0.9887	0.9966	0.9962	0.9951	0.9938	0.9951	0.9923	0.9997	0.9996	0.9998	0.9997	0.9995	0.9996	0.9998	0.9998	0.9941	0.9934	0.9932	0.9935
	Max Error	0.2131	1.3215	0.7718	0.9155	0.8919	0.8806	0.4144	0.4426	0.9734	2.5527	1.269	0.8342	0.1146	0.4993	0.263	0.3684	0.335	0.2821	0.1162	0.0993	1.1694	1.2374	1.1625	0.9813
	MAE	0.0573	0.5297	0.2655	0.3245	0.2919	0.2719	0.1519	0.1562	0.4831	1.4098	0.6802	0.4152	0.0643	0.3177	0.1707	0.2463	0.2515	0.2067	0.0391	0.0533	0.7487	0.7913	0.7515	0.6149
	AUC	24.6676	35.9616	29.6529	31.0587	30.2515	29.745	26.9763	27.0835	34.9723	57.3323	39.7287	33.3213	45.422	54.2442	50.6675	52.5135	52.687	51.5744	34.9521	34.5683	54.0525	55.096	54.1411	50.8177
	SNR	23.6538	5.4053	10.9487	9.2434	9.9627	10.4201	16.428	15.8983	6.7983	-2.5319	3.9787	8.2175	30.7098	16.903	22.4264	19.2061	19.2406	20.9042	31.667	29.5241	6.7775	6.2788	6.7774	8.454

b) Scenario: Lane Change in front of the Lead car on US101 Highway - Original																									
Trajectory Prediction Model		CS-LSTM											STDAN												
Manouver		Right Lane change						Left Lane change						Right Lane change						Left Lane change					
Sensor Parameter		HFOV						HFOV						HFOV						HFOV					
Sensor Modality	Comparison Metrics	30deg	60deg	90deg	120deg	150deg	180deg	30deg	60deg	90deg	120deg	150deg	180deg	30deg	60deg	90deg	120deg	150deg	180deg	30deg	60deg	90deg	120deg	150deg	180deg
Radar	Pearson Coeff	0.9958	0.9913	0.9895	0.9858	0.9839	0.9912	0.9961	0.9951	0.9931	0.9889	0.9905	0.9911	0.9993	0.9968	0.9972	0.9973	0.9976	0.9975	0.9993	0.9994	0.9918	0.9913	0.9921	0.992
	Max Error	0.4202	2.0786	0.9912	1.2761	1.4814	1.8458	0.4851	0.5237	1.5157	1.939	1.9938	2.0774	0.1811	1.6788	1.5202	1.4066	1.542	1.5934	0.5336	0.4278	1.4798	1.5052	1.5179	1.4581
	MAE	0.1667	1.1387	0.3953	0.4646	0.535	1.0058	0.218	0.2408	0.8611	1.1966	1.2065	1.2775	0.1018	1.008	0.9343	0.873	0.9318	0.9545	0.1702	0.1422	0.8775	0.9007	0.8885	0.8951
	AUC	27.2131	50.6329	32.6085	34.235	35.8789	47.4198	28.2984	28.8809	43.8184	51.9918	52.2488	53.9038	37.5332	59.5412	57.7341	56.2567	57.6574	58.2064	35.5234	36.9047	60.6368	61.207	60.8882	61.0779
	SNR	15.4727	-0.7239	7.9415	6.0491	4.8702	0.3688	13.4619	12.557	1.9426	-0.9789	-1.0772	-1.5063	24.2544	4.2483	4.9882	5.5775	4.9458	4.7397	18.3193	20.3779	6.2798	6.0633	6.1462	6.1875

c) Scenario: Deceleration on US101 Highway - Original															
Trajectory Prediction Model		CS-LSTM						STDAN							
Manouver		Deceleration						Deceleration							
Sensor Parameter		HFOV						HFOV							
Sensor Modality	Comparison Metrics	30deg	60deg	90deg	120deg	150deg	180deg	30deg	60deg	90deg	120deg	150deg	180deg	30deg	60deg
Radar	Pearson Coeff	0.996	0.9965	0.996	0.9968	0.9969	0.997	0.9995	0.9995	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
	Spearman Coeff	0.9977	0.9977	0.9977	0.9977	0.9977	0.9992	0.9992	0.9992	0.9985	0.9985	0.9985	0.9985	0.9985	0.9985
	MAE	0.1086	0.0687	0.0958	0.0662	0.0642	0.042	0.1582	0.1632	0.1366	0.1359	0.1349	0.1396	0.1396	0.1396
	AUC	26.4702	25.4455	26.155	25.421	25.3519	23.9971	31.1725	31.0542	38.1586	38.1418	38.1163	38.229	38.229	38.229
	SNR	18.8695	22.4608	20.0356	22.8155	23.135	27.1354	19.6773	19.3877	21.2058	21.2545	21.2995	21.0226	21.0226	21.0226

Angular Resolution

a) Scenario: Lane Change in front of the Ego car on US101 Highway - Original

Trajectory Prediction Model		CS-LSTM										STDAN									
Manouver		Right Lane change					Left Lane change					Right Lane change					Left Lane change				
Sensor Parameter		Angular Resolution					Angular Resolution					Angular Resolution					Angular Resolution				
Sensor Modality	Comparison Metrics	2deg	4deg	6deg	8deg	10deg	2deg	4deg	6deg	8deg	10deg	2deg	4deg	6deg	8deg	10deg	2deg	4deg	6deg	8deg	10deg
Radar	Pearson Coeff	0.9881	0.9882	0.9864	0.99	0.9918	0.9935	0.9892	0.9919	0.993	0.993	0.9995	0.9993	0.9996	0.9998	0.9998	0.9967	0.9977	0.9943	0.9935	0.9963
	Max Error	0.9384	1.3349	1.6498	1.4401	1.1971	1.5847	2.4383	2.7322	3.1405	3.2981	0.3356	0.3457	0.3002	0.1316	0.1216	1.0389	1.0616	1.3472	1.3167	1.5005
	MAE	0.301	0.5217	0.6572	0.6277	0.4963	0.8751	1.0815	1.5538	1.6905	1.7747	0.2545	0.2711	0.2245	0.0731	0.0618	0.6722	0.6445	0.8592	0.8362	0.9171
	AUC	30.458	35.7681	38.9889	38.3556	35.1963	44.4136	49.1696	60.8229	64.029	66.048	52.7573	53.1722	52.0148	45.1743	45.4812	52.1422	51.4253	56.7138	56.1785	58.074
	SNR	9.7121	5.4571	3.5524	4.0847	6.0889	1.7115	-0.5931	-3.3472	-4.147	-4.6042	19.1461	18.6102	20.2188	29.3956	30.9084	7.7317	7.8878	5.5346	5.7714	4.857

b) Scenario: Lane Change in front of the Lead car on US101 Highway - Original

Trajectory Prediction Model		CS-LSTM										STDAN									
Manouver		Right Lane change					Left Lane change					Right Lane change					Left Lane change				
Sensor Parameter		Angular Resolution					Angular Resolution					Angular Resolution					Angular Resolution				
Sensor Modality	Comparison Metrics	2deg	4deg	6deg	8deg	10deg	2deg	4deg	6deg	8deg	10deg	2deg	4deg	6deg	8deg	10deg	2deg	4deg	6deg	8deg	10deg
Radar	Pearson Coeff	0.9881	0.9908	0.9927	0.9876	0.9913	0.9923	0.9936	0.9924	0.9928	0.992	0.9976	0.9972	0.9992	0.9975	0.9991	0.9925	0.9996	0.995	0.9934	0.9992
	Max Error	1.0812	1.2506	0.7965	1.6021	0.9986	2.2234	0.7133	3.2296	3.0151	1.4125	1.4949	1.5973	0.2635	1.5546	0.2556	1.6071	0.5218	1.6744	1.6359	0.2729
	MAE	0.4319	0.596	0.3253	0.6852	0.4231	1.2675	0.3137	1.6941	1.5842	0.6317	0.9204	0.9809	0.145	0.9518	0.1449	0.9468	0.1803	0.9893	0.9683	0.1016
	AUC	33.4689	37.5224	30.9866	39.5597	33.3203	53.623	30.5716	63.748	61.1393	38.1764	57.3808	58.8594	38.5487	58.1392	38.551	62.2782	35.2626	63.2521	62.777	39.1559
	SNR	7.2129	4.728	9.6042	3.2628	7.2926	-1.6149	10.1479	-4.2765	-3.7252	3.9202	5.1191	4.5644	21.0944	4.8226	21.1617	5.5668	18.0968	5.1373	5.3711	24.3142

c) Scenario: Deceleration on US101 Highway - Original

Trajectory Prediction Model		CS-LSTM					STDAN				
Manouver		Deceleration					Deceleration				
Sensor Parameter		Angular Resolution					Angular Resolution				
Sensor Modality	Comparison Metrics	2deg	4deg	6deg	8deg	10deg	2deg	4deg	6deg	8deg	10deg
Radar	Pearson Coeff	0.9968	0.9971	0.9955	0.9965	0.9964	0.9989	0.9988	0.9988	1	1
	Max Error	0.1806	0.2864	0.3937	0.3707	0.3801	0.634	0.6291	0.6155	0.0793	0.0706
	MAE	0.0438	0.0828	0.1119	0.1199	0.1446	0.3459	0.3447	0.3375	0.0541	0.0516
	AUC	24.3086	25.8627	26.5241	26.7421	27.3519	38.0894	38.063	37.8883	31.0777	31.0192
	SNR	26.3872	21.2034	18.5866	18.4583	17.2692	11.8402	11.881	12.0736	28.7201	29.2062